

Part 1:

1. Top module

```
1 import uvm_pkg::*;
2 `include "uvm_macros.svh"
3 import alsu_test_pkg::*;
4
5 module top();
6
7     bit clk;
8
9     initial begin
10         clk = 0;
11         forever
12             #1 clk = ~clk;
13     end
14
15     alsu_if alsuif (clk);
16
17     ALSU DUT (.clk(clk), .A(alsuif.A), .B(alsuif.B),
18             .cin(alsuif.cin), .serial_in(alsuif.serial_in),
19             .red_op_A(alsuif.red_op_A), .red_op_B(alsuif.red_op_B),
20             .opcode(alsuif.opcode), .bypass_A(alsuif.bypass_A),
21             .bypass_B(alsuif.bypass_B), .rst(alsuif.rst), .direction(alsuif.direction),
22             .leds(alsuif.leds), .out(alsuif.out));
23
24     initial begin
25         run_test("alsu_test");
26     end
27
28 endmodule
```

2. ALSU Interface

```
1 interface alsu_if(clk);
2     input clk;
3     logic cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
4     logic [2:0] opcode;
5     logic signed [2:0] A, B;
6     logic [15:0] leds;
7     logic signed [5:0] out;
8 endinterface : alsu_if
```

3. Alsu test package

```
1 package alsu_test_pkg;
2
3     import uvm_pkg::*;
4     import alsu_env_pkg::*;
5     `include "uvm_macros.svh"
6
7
8     class alsu_test extends uvm_test;
9         `uvm_component_utils(alsu_test)
10
11         alsu_env env;
12
13         function new(string name = "alsu_test", uvm_component parent = null);
14             super.new(name, parent);
15         endfunction
16
17         function void build_phase(uvm_phase phase);
18             super.build_phase(phase);
19             env = alsu_env::type_id::create("env", this);
20         endfunction
21
22         task run_phase(uvm_phase phase);
23             super.run_phase(phase);
24             phase.raise_objection(this);
25             #100; `uvm_info("run_phase", "Inside the ALSU test", UVM_MEDIUM);
26             phase.drop_objection(this);
27         endtask
28     endclass
29
30 endpackage
```

4. Alsu Environment

```
1 package alsu_env_pkg;
2
3 import uvm_pkg::*;
4 `include "uvm_macros.svh"
5
6 class alsu_env extends uvm_env;
7     `uvm_component_utils(alsu_env);
8
9     function new(string name = "alsu_env", uvm_component parent = null);
10         super.new(name, parent);
11     endfunction
12
13 endclass
14
15 endpackage
```

5. Questasim output

```
***** IMPORTANT RELEASE NOTES *****
#
# You are using a version of the UVM library that has been compiled
# with `UVM_NO_DEPRECATED undefined.
# See http://www.eda.org/svdb/view.php?id=3313 for more details.
#
# You are using a version of the UVM library that has been compiled
# with `UVM_OBJECT_MUST_HAVE_CONSTRUCTOR undefined.
# See http://www.eda.org/svdb/view.php?id=3770 for more details.
#
# (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test alsu_test...
# UVM_INFO alsu_test.sv(25) @ 100: uvm_test_top [run_phase] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 5
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
#
# ** Report counts by id
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [run_phase] 1
#
# ** Note: $finish : D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 100 ns Iteration: 54 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
```

Test info message

Part 2

1. Top Module

```
1  import uvm_pkg::*;
2  `include "uvm_macros.svh"
3  import alsu_test_pkg::*;
4
5  module top();
6
7      bit clk;
8
9      initial begin
10         clk = 0;
11         forever
12             #1 clk = ~clk;
13     end
14
15     alsu_if alsuif (clk);
16
17     ALSU DUT (.clk(clk), .A(alsuif.A), .B(alsuif.B),
18             .cin(alsuif.cin), .serial_in(alsuif.serial_in),
19             .red_op_A(alsuif.red_op_A), .red_op_B(alsuif.red_op_B),
20             .opcode(alsuif.opcode), .bypass_A(alsuif.bypass_A),
21             .bypass_B(alsuif.bypass_B), .rst(alsuif.rst), .direction(alsuif.direction),
22             .leds(alsuif.leds), .out(alsuif.out));
23
24     initial begin
25         uvm_config_db#(virtual alsu_if)::set(null, "uvm_test_top", "vif", alsuif);
26         run_test("alsu_test");
27     end
28
29 endmodule
```

2. ALSU Interface

```
1  interface alsu_if(clk);
2      input clk;
3      logic cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
4      logic [2:0] opcode;
5      logic signed [2:0] A, B;
6      logic [15:0] leds;
7      logic signed [5:0] out;
8  endinterface : alsu_if
```

3. ALSU test

```
1 package alsu_test_pkg;
2
3 import uvm_pkg::*;
4 import alsu_env_pkg::*;
5 `include "uvm_macros.svh"
6
7
8 class alsu_test extends uvm_test;
9     `uvm_component_utils(alsu_test)
10
11     alsu_env env;
12     virtual alsu_if alsu_test_vif;
13
14     function new(string name = "alsu_test", uvm_component parent = null);
15         super.new(name, parent);
16     endfunction
17
18     function void build_phase(uvm_phase phase);
19         super.build_phase(phase);
20         env = alsu_env::type_id::create("env", this);
21         if(!uvm_config_db #(virtual alsu_if)::get(this, "", "vif", alsu_test_vif))
22             `uvm_fatal("build_phase", "Test - Unable to get Alsu Virtual Interface");
23
24         uvm_config_db #(virtual alsu_if)::set(null, "*", "test_vif", alsu_test_vif);
25     endfunction
26
27     task run_phase(uvm_phase phase);
28         super.run_phase(phase);
29         phase.raise_objection(this);
30         #100; `uvm_info("run_phase", "Inside the ALSU test", UVM_MEDIUM);
31         phase.drop_objection(this);
32     endtask
33 endclass
34
35 endpackage
```

4. ALSU Environment



```
1  package alsu_env_pkg;
2
3      import uvm_pkg::*;
4      import alsu_driver_pkg::*;
5      `include "uvm_macros.svh"
6
7      class alsu_env extends uvm_env;
8          `uvm_component_utils(alsu_env);
9
10         alsu_driver driver;
11
12         function new(string name = "alsu_env", uvm_component parent = null);
13             super.new(name, parent);
14             driver = alsu_driver::type_id::create("driver", this);
15         endfunction
16
17     endclass
18
19 endpackage
```

5. ALSU Driver

```
1  package alsu_driver_pkg;
2
3      import uvm_pkg::*;
4      `include "uvm_macros.svh"
5
6      class alsu_driver extends uvm_driver;
7          `uvm_component_utils(alsu_driver);
8
9          virtual alsu_if alsu_driver_vif;
10
11          function new(string name = "alsu_driver", uvm_component parent);
12              super.new(name, parent);
13          endfunction
14
15          function void build_phase(uvm_phase phase);
16              super.build_phase(phase);
17              uvm_config_db #(virtual alsu_if)::get(this, "*", "test_vif", alsu_driver_vif);
18          endfunction
19
20          task run_phase(uvm_phase phase);
21              super.run_phase(phase);
22              alsu_driver_vif.rst = 1;
23              alsu_driver_vif.cin = 0;
24              alsu_driver_vif.red_op_A = 0;
25              alsu_driver_vif.red_op_B = 0;
26              alsu_driver_vif.bypass_A = 0;
27              alsu_driver_vif.bypass_B = 0;
28              alsu_driver_vif.direction = 0;
29              alsu_driver_vif.serial_in = 0;
30              alsu_driver_vif.opcode = 3'b000;
31              alsu_driver_vif.A = 3'b000;
32              alsu_driver_vif.B = 3'b000;
33              @(negedge alsu_driver_vif.clk);
34              forever begin
35                  alsu_driver_vif.rst = 0;
36                  alsu_driver_vif.cin = $random;
37                  alsu_driver_vif.red_op_A = $random;
38                  alsu_driver_vif.red_op_B = $random;
39                  alsu_driver_vif.bypass_A = $random;
40                  alsu_driver_vif.bypass_B = $random;
41                  alsu_driver_vif.direction = $random;
42                  alsu_driver_vif.serial_in = $random;
43                  alsu_driver_vif.opcode = $random;
44                  alsu_driver_vif.A = $random;
45                  alsu_driver_vif.B = $random;
46                  @(negedge alsu_driver_vif.clk);
47              end
48          endtask
49      endclass
50
51  endpackage
```

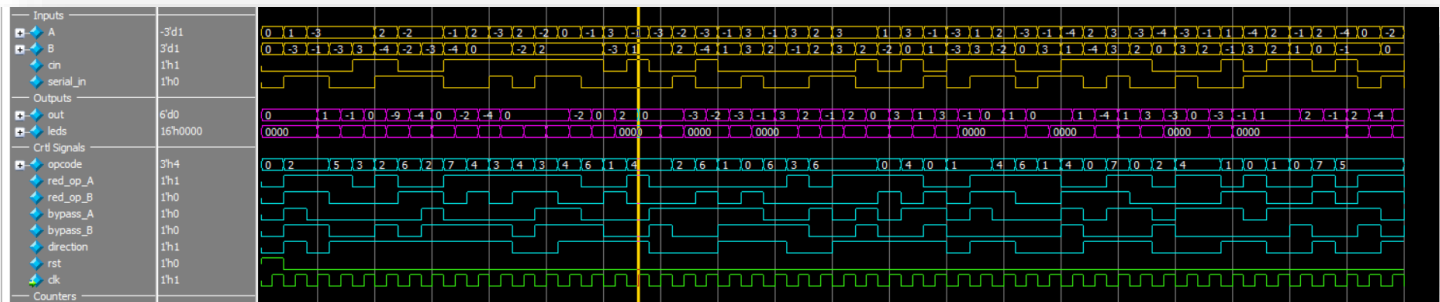
6. Questasim Output

```

# ***** IMPORTANT RELEASE NOTES *****
#
# You are using a version of the UVM library that has been compiled
# with `UVM_NO_DEPRECATED undefined.
# See http://www.eda.org/svdb/view.php?id=3313 for more details.
#
# You are using a version of the UVM library that has been compiled
# with `UVM_OBJECT_MUST_HAVE_CONSTRUCTOR undefined.
# See http://www.eda.org/svdb/view.php?id=3770 for more details.
#
# (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test alsu_test...
# UVM_INFO alsu_test.sv(30) @ 100: uvm_test_top [run_phase] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 5
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
#
# ** Report counts by id
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [run_phase] 1
#
# ** Note: $finish : D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 100 ns Iteration: 54 Instance: /top
#
# Break in Task uvm_pkg/uvm_root::run_test at D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
VSIM(paused)>

```

7. Questasim Waveform



Part 3

1. Top Module

```
1  import uvm_pkg::*;
2  `include "uvm_macros.svh"
3  import alsu_test_pkg::*;
4
5  module top();
6
7      bit clk;
8
9      initial begin
10         clk = 0;
11         forever
12             #1 clk = ~clk;
13     end
14
15     alsu_if alsuif (clk);
16
17     ALSU DUT (.clk(clk), .A(alsuif.A), .B(alsuif.B),
18             .cin(alsuif.cin), .serial_in(alsuif.serial_in),
19             .red_op_A(alsuif.red_op_A), .red_op_B(alsuif.red_op_B),
20             .opcode(alsuif.opcode), .bypass_A(alsuif.bypass_A),
21             .bypass_B(alsuif.bypass_B), .rst(alsuif.rst), .direction(alsuif.direction),
22             .leds(alsuif.leds), .out(alsuif.out));
23
24     initial begin
25         uvm_config_db#(virtual alsu_if)::set(null, "uvm_test_top", "vif", alsuif);
26         run_test("alsu_test");
27     end
28
29 endmodule
```

2. ALSU Interface

```
1  interface alsu_if(clk);
2      input clk;
3      logic cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
4      logic [2:0] opcode;
5      logic signed [2:0] A, B;
6      logic [15:0] leds;
7      logic signed [5:0] out;
8  endinterface : alsu_if
```

3. ALSU Test

```
1 package alsu_test_pkg;
2
3 import uvm_pkg::*;
4 import alsu_env_pkg::*;
5 import alsu_config_pkg::*;
6 `include "uvm_macros.svh"
7
8
9 class alsu_test extends uvm_test;
10     `uvm_component_utils(alsu_test)
11
12     alsu_env env;
13     alsu_config alsu_config_obj_test;
14
15     function new(string name = "alsu_test", uvm_component parent = null);
16         super.new(name, parent);
17     endfunction
18
19     function void build_phase(uvm_phase phase);
20         super.build_phase(phase);
21         env = alsu_env::type_id::create("env", this);
22         alsu_config_obj_test = alsu_config::type_id::create("alsu_config_obj_test");
23         if(!uvm_config_db #(virtual alsu_if)::get(this, "", "vif", alsu_config_obj_test.alsu_config_vif))
24             `uvm_fatal("build_phase", "Test - Unable to get Alsu Virtual Interface");
25
26         uvm_config_db #(alsu_config)::set(null, "*", "CFG", alsu_config_obj_test);
27     endfunction
28
29     task run_phase(uvm_phase phase);
30         super.run_phase(phase);
31         phase.raise_objection(this);
32         #100; `uvm_info("run_phase", "Inside the ALSU test", UVM_MEDIUM);
33         phase.drop_objection(this);
34     endtask
35 endclass
36
37 endpackage
```

4. ALSU Environment

```
1 package alsu_env_pkg;
2
3     import uvm_pkg::*;
4     import alsu_driver_pkg::*;
5     `include "uvm_macros.svh"
6
7     class alsu_env extends uvm_env;
8         `uvm_component_utils(alsu_env);
9
10        alsu_driver driver;
11
12        function new(string name = "alsu_env", uvm_component parent = null);
13            super.new(name, parent);
14            driver = alsu_driver::type_id::create("driver", this);
15        endfunction
16
17    endclass
18
19 endpackage
```

5. ALSU Config

```
1 package alsu_config_pkg;
2     import uvm_pkg::*;
3     `include "uvm_macros.svh"
4
5     class alsu_config extends uvm_object;
6         `uvm_object_utils(alsu_config)
7
8         // Configuration parameters
9         virtual alsu_if alsu_config_vif;
10
11        function new(string name = "alsu_config");
12            super.new(name);
13        endfunction
14
15
16    endclass : alsu_config
17 endpackage
```

6. ALSU Driver

```
1 package alsu_driver_pkg;
2
3 import uvm_pkg::*;
4 import alsu_config_pkg::*;
5 `include "uvm_macros.svh"
6
7 class alsu_driver extends uvm_driver;
8     `uvm_component_utils(alsu_driver);
9
10     virtual alsu_if alsu_driver_vif;
11     alsu_config alsu_config_obj_test;
12
13
14     function new(string name = "alsu_driver", uvm_component parent);
15         super.new(name, parent);
16     endfunction
17
18     function void build_phase(uvm_phase phase);
19         super.build_phase(phase);
20         if (!uvm_config_db #(alsu_config)::get(this, "", "CFG", alsu_config_obj_test))
21             `uvm_fatal("build_phase", "Driver - alsu_config_obj_test was not found in the uvm_config_db")
22         endfunction
23
24     function void connect_phase(uvm_phase phase);
25         super.connect_phase(phase);
26         alsu_driver_vif = alsu_config_obj_test.alsu_config_vif;
27     endfunction
28
29     task run_phase(uvm_phase phase);
30         super.run_phase(phase);
31         alsu_driver_vif.rst = 1;
32         alsu_driver_vif.cin = 0;
33         alsu_driver_vif.red_op_A = 0;
34         alsu_driver_vif.red_op_B = 0;
35         alsu_driver_vif.bypass_A = 0;
36         alsu_driver_vif.bypass_B = 0;
37         alsu_driver_vif.direction = 0;
38         alsu_driver_vif.serial_in = 0;
39         alsu_driver_vif.opcode = 3'b000;
40         alsu_driver_vif.A = 3'b000;
41         alsu_driver_vif.B = 3'b000;
42         @(negedge alsu_driver_vif.clk);
43         forever begin
44             alsu_driver_vif.rst = 0;
45             alsu_driver_vif.cin = $random;
46             alsu_driver_vif.red_op_A = $random;
47             alsu_driver_vif.red_op_B = $random;
48             alsu_driver_vif.bypass_A = $random;
49             alsu_driver_vif.bypass_B = $random;
50             alsu_driver_vif.direction = $random;
51             alsu_driver_vif.serial_in = $random;
52             alsu_driver_vif.opcode = $random;
53             alsu_driver_vif.A = $random;
54             alsu_driver_vif.B = $random;
55             @(negedge alsu_driver_vif.clk);
56         end
57     endtask
58 endclass
59
60 endpackage
```

7. Questasim Output

```
#-----
#
# ***** IMPORTANT RELEASE NOTES *****
#
# You are using a version of the UVM library that has been compiled
# with `UVM_NO_DEPRECATED undefined.
# See http://www.eda.org/svdb/view.php?id=3313 for more details.
#
# You are using a version of the UVM library that has been compiled
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#
# UVM_INFO verililog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verililog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTSI] Running test alsu_test...
# UVM_INFO alsu_test.sv(32) @ 100: uvm_test_top [run_phase] Inside the ALSU test
# UVM_INFO verililog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 5
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
#
# ** Report counts by id
# [Questa UVM] 2
# [RNTSI] 1
# [TEST_DONE] 1
# [run_phase] 1
#
# ** Note: $finish : D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 100 ns Iteration: 54 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/Mina/Program_Files/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
#
# V$IM(pausd)>
```

8. Questasim Waveform

